

tests/dynamic/suites/ benchmark_suite.sh — operator guide

Revision: 1 **Last modified:** 2026-07-01T00:00:00Z **Status:**
Authored for P10. Honest-SKIPs (topology_unsupported, §11.4.69)
until the live dynamic stack lands; parse-clean + authored today.

Companion (§11.4.18) to the in-source documentation block
at the top of the script.

Overview

§11.4.169 **performance / benchmark** suite. It measures per-request latency through the live dynamic stack, computes **p50/p95/p99** from real captured per-request times, and asserts (a) p95 is within an absolute **budget** and (b) — when a recorded baseline exists — there is **no regression** beyond a tolerance vs that baseline. It records the full latency series + percentiles as the captured evidence (§11.4.69); a regression vs baseline is a finding (§11.4.169 benchmarking clause).

Prerequisites

- Source library tests/dynamic/lib/analyzer_common.sh + committed tests/lib/evidence.sh.
- curl, awk, sort, POSIX sh, date.
- For a real run: HELIX_DYNAMIC_STACK=1 + HELIX_PROXY_URL.

Usage examples

```
# Today (no live stack): honest SKIP, exit 0:  
bash tests/dynamic/suites/benchmark_suite.sh
```

```
# P10 live run:  
HELIX_DYNAMIC_STACK=1 HELIX_PROXY_URL=http://127.0.0.1:53128 \  
  BENCH_N=200 BENCH_P95_BUDGET_MS=800 BENCH_REGRESS_PCT=25 \  
bash tests/dynamic/suites/benchmark_suite.sh
```

Env: BENCH_N (samples, default 200), BENCH_TARGET (default `http://target-a.internal/`), BENCH_P95_BUDGET_MS (default 800), BENCH_REGRESS_PCT (max allowed p95 growth vs baseline, default 25), BENCH_BASELINE (default `qa-results/p9-harness/bench_baseline.p95`).

Edge cases

- **Live stack absent** → `topology_unsupported SKIP`, exit 0.
- **No baseline file yet** → budget-only check; a passing run **writes** the baseline so future runs gain the regression check.
- **p95 over budget OR regression > BENCH_REGRESS_PCT** vs the baseline → `FAIL` citing `benchmark.evidence`.

§11.4.115 RED_MODE polarity

RED_MODE	What it asserts	PASS means
0 (default)	GREEN guard	p95 ≤ budget AND no regression beyond tolerance
1	RED baseline	the pre-fix stack's p95 exceeds budget (regression) → defect reproduced

A RED_MODE=1 run where p95 stayed within budget is a finding (no regression to reproduce), not a pass.

Internal behaviour

- `#!/usr/bin/env bash, POSIX-clean (sh -n + bash -n, §11.4.67).`
- Captures `BENCH_N` `%{time_total}` samples (converted to integer ms via `awk`) into `latency_ms.series`; nearest-rank `awk` computes p50/p95/p99.
- Reads the baseline p95 (if any), computes the regression %, writes `benchmark.evidence` (percentiles + series). GREEN requires within-budget AND no-regression; on pass it refreshes the baseline file. Evidence dir: `qa-results/p9-harness/benchmark_<run-id>/.`
- Resources: shell+curl only; bounded sample count; §12.6 60% host ceiling.

Related

- tests/dynamic/lib/analyzer_common.sh — sourced base.
- tests/dynamic/suites/run_all_suites.sh — orchestrates this suite.
- Constitution §11.4.169 / §11.4.24 / §11.4.50 / §11.4.69 / §11.4.115 / §12.6; design spec §13.

Last verified

2026-07-01 — run with no live stack: honest SKIP, exit 0; sh -n + bash -n parse-clean. Live latency measurement + baseline regression check are exercised in **P10**.